CHEMISTRY — SOLUTIONS

Chapter 1 | Objective Practice Paper | Class 12 | Maximum Time: 120 Minutes

SECTION A — MULTIPLE CHOICE QUESTIONS (100 MCQS)

1. For which of the following, the depression in freezing point will be maximum?
(A) K_2SO_4 (B) $NaCl$ (C) Urea (D) Glucose

2. Which of the following is NOT a colligative property?
(A) Depression in freezing point (B) Optical activity
(C) Relative lowering of vapour pressure (D) Elevation in boiling point

3. In 234.2 g of sugar syrup, 34.2 g of sugar is present. What will be the molal concentration of the solution?
(A) 0.1 m (B) 0.5 m (C) 5.5 m (D) 0.25 m

4. A 5% cane sugar solution (molar mass = 342) is isotonic with a 1% solution of an unknown solute X. The molar mass of X is:
(A) 34.2 (B) 171.2 (C) 68.4 (D) 136.8

5. 2 g of NaOH is dissolved in 200 mL of solution. The molarity of the solution will be:
(A) 0.25 M (B) 0.5 M (C) 5 M (D) 0.1 M

6. Which of the following concentration units remains unaffected by a change in temperature?
(A) Molarity (B) Normality (C) Molality (D) Formality

7. The composition of the azeotropic mixture of HCl and H_2O is:
(A) 48% HCl (B) 22.2% HCl
(C) 36% HCl (D) 20.2% HCl

8. An aqueous solution has an osmotic pressure of 0.0821 atm at 300 K. Its concentration will be:
(A) 0.033 M (B) 0.0033 M (C) 0.33 M (D) 0.082 M

9. The van 't Hoff factor of 0.1 M $Ba(NO_3)_2$ solution is 2.74. The degree of dissociation is:
(A) 91.3% (B) 87% (C) 100% (D) 74%

10. Soda water is an example of which of the following?
(A) Solution of CO_2 gas in water
(B) Solution of gas in gas
(C) Solution of liquid in solid
(D) Solution of solid in liquid

11. In comparison to 0.01 M glucose solution, the depression in freezing point of 0.01 M $MgCl_2$ solution will be:
(A) Same (B) About twice (C) About thrice (D) About six times

12. Upon complete dissociation of $Ca(NO_3)_2$, what is the value of the van 't Hoff factor (i)?
(A) 1 (B) 2 (C) 3 (D) 4

13. The number of moles of solute present in 1 kg of solvent is called:
(A) Molarity (B) Molality (C) Normality (D) Mole fraction

14. A liquid mixture which boils at a constant temperature without any change in composition is called:
(A) Ideal solution (B) Supersaturated solution
(C) Azeotropic mixture (D) Unsaturated solution

15. Which of the following is NOT an example of an ideal solution?
(A) Benzene + Toluene (B) n-Hexane + n-Heptane
(C) Chloroform + Acetone (D) Chloroethane + Bromoethane

16. During the process of osmosis, what passes through a semipermeable membrane?
(A) Only solvent molecules (B) Only solute molecules
(C) Complex ions (D) Both solute and solvent molecules

17. If 18 g of glucose ($C_6H_{12}O_6$) is dissolved in 1000 g of solvent, the solution is said to be:
(A) 1 molar (B) 0.1 molal (C) 0.1 molar (D) 0.5 molal

18. Which of the following aqueous solutions will have the highest boiling point?
(A) 1% glucose (B) 1% sucrose (C) 1% $NaCl$ (D) 1% urea

19. At a given temperature, isotonic solutions always possess identical:
(A) Vapour pressure (B) Density
(C) Molar concentration (D) Surface tension

20. Which of the following liquid pairs shows a positive deviation from Raoult's law?
(A) Benzene + Toluene (B) Ethanol + Acetone
(C) Chloroform + Acetone (D) Phenol + Aniline

21. The osmotic pressure (Π) of a dilute solution is correctly represented by:
(A) $\Pi = \frac{C}{RT}$ (B) $\Pi = CRT$ (C) $\Pi = \frac{RT}{C}$ (D) $\Pi = \frac{CT}{R}$

22. The molarity of pure water at room temperature is approximately:
(A) 18 M (B) 55.55 M (C) 50 M (D) 1 M

23. Who first gave a quantitative relationship between the solubility of a gas in a liquid and pressure?
(A) Raoult (B) Henry (C) Dalton (D) van 't Hoff

24. With an increase in temperature, the value of Henry's constant (K_H):
(A) Increases (B) Decreases
(C) Remains unchanged (D) First decreases then increases

25. Aquatic species are more comfortable in cold water rather than warm water because:
(A) Solubility of gases is lower in cold water
(B) Solubility of oxygen is higher in cold water
(C) K_H value is lower in warm water
(D) Surface tension of cold water is higher

26. Why is helium added to breathing air cylinders used by deep-sea scuba divers?
(A) It makes oxygen heavier
(B) Its solubility in blood is extremely low
(C) It induces toxicity
(D) It contracts lung alveoli

27. Weakness and inability to think clearly (anoxia) experienced by climbers at high altitudes is caused by:
(A) Low atmospheric temperature
(B) High atmospheric pressure
(C) Low partial pressure of oxygen
(D) Excess carbon dioxide in blood

28. The enthalpy of mixing ($\Delta_{mix}H$) during the formation of an ideal solution is:
(A) Positive (B) Negative (C) Zero (D) Infinite

29. The volume change of mixing ($\Delta_{mix}V$) for an ideal solution is:
(A) Zero (B) Positive (C) Negative (D) Variable

30. When mixing liquids A and B, if A-B intermolecular forces are weaker than A-A and B-B interactions, the solution shows:
(A) Positive deviation from Raoult's law
(B) Negative deviation from Raoult's law
(C) Perfectly ideal behavior
(D) Complete insolubility

31. A mixture of acetone and chloroform shows negative deviation from Raoult's law due to:
(A) Ionic bond formation
(B) Formation of hydrogen bonding between molecules
(C) Covalent bonding
(D) Sharp increase in vapour pressure

32. A mixture of water and nitric acid containing 68% by mass forms which type of azeotrope?
(A) Minimum boiling azeotrope (B) Maximum boiling azeotrope
(C) Ideal mixture (D) Heterogeneous mixture

33. Fractional distillation of an ethanol-water mixture can yield ethanol up to a maximum concentration of:
(A) 100% (B) 95% (C) 68% (D) 50%

34. The relative lowering of vapour pressure is equal to which property of the non-volatile solute?
(A) Molarity (B) Molality (C) Mole fraction (D) Mass percentage

35. The unit of the molal elevation constant (K_b) is:
(A) K kg mol^{-1} (B) K mol kg^{-1} (C) kg mol^{-1} (D) K mol^{-1}

36. The value of the molal depression constant (K_f) for water is:
(A) 0.52 (B) 1.86 (C) 5.12 (D) 3.90

37. The value of the molal elevation constant (K_b) for water is:
(A) 1.86 (B) 0.52 (C) 2.53 (D) 0.99

38. The primary role of a 35% (v/v) ethylene glycol solution added to car radiators is:
(A) To heat the engine (B) To lower the freezing point of water
(C) To reduce the boiling point of water (D) To increase engine friction

39. When red blood cells are placed in a $NaCl$ solution containing more than 0.9% (m/V), they will:
(A) Swell (B) Shrink
(C) Remain unaffected (D) Burst immediately

40. Which process is utilized to obtain pure drinking water from saline seawater?
(A) Osmosis (B) Reverse osmosis
(C) Electrolysis (D) Coagulation

41. The commercial semipermeable membrane widely used in reverse osmosis desalination is made of:
(A) Cellulose acetate (B) Teflon
(C) Bakelite (D) Nylon

42. Due to the dimerization of acetic acid in benzene, the value of the van 't Hoff factor (i) is:
(A) 1 (B) 2 (C) Approximately 0.5 (D) 0

43. When a solute undergoes complete association in solution, its van 't Hoff factor (i) satisfies:
(A) $i = 1$ (B) $i > 1$ (C) $i < 1$ (D) $i = 0$

44. When a solute undergoes complete dissociation in solution, the value of i satisfies:
(A) $i > 1$ (B) $i < 1$ (C) $i = 1$ (D) $i = 0$

45. Upon complete dissociation of $K_4[Fe(CN)_6]$, the value of the van 't Hoff factor (i) will be:
(A) 2 (B) 3 (C) 4 (D) 5

46. In a binary solution, the sum of the mole fractions of the solute and the solvent is always:
(A) 0 (B) 1 (C) 2 (D) 100

47. Osmotic pressure measurement is preferred for determining the molar masses of biomolecules because:
(A) It must be performed at very high temperatures
(B) Its magnitude is sufficiently large even at room temperature
(C) It is valid only for inorganic salts
(D) It does not require any liquid solvent

48. When a non-volatile solute is added to a pure solvent, its vapour pressure:
(A) Increases (B) Decreases
(C) Remains unchanged (D) Drops to zero

49. Raw mangoes shrivel when placed in concentrated brine solution due to:
(A) Endosmosis (B) Exosmosis
(C) Diffusion (D) Evaporation

50. Swelling and puffiness in tissue cells caused by water retention in people who consume excess salt is termed:
(A) Edema (B) Anoxia (C) Bends (D) Ulcer

51. One litre of seawater contains approximately 6 mg of dissolved oxygen. In parts per million (ppm), this is:
(A) 0.58 ppm (B) 5.8 ppm (C) 58 ppm (D) 580 ppm

52. A mixture of ethyl alcohol and water is an example of a:
(A) Liquid in liquid solution (B) Solid in liquid solution
(C) Gas in liquid solution (D) Solid in solid solution

53. Brass is a homogeneous solid solution of which metals?
(A) Cu and Sn (B) Cu and Zn (C) Cu, Zn and Ni (D) Fe and C

54. German silver is an alloy composed of:
(A) Copper and tin (B) Copper, zinc and nickel
(C) Copper and silver (D) Lead and tin

55. Bronze is an example of an alloy composed of:
(A) Cu and Zn (B) Cu and Sn (C) Fe and Cr (D) Pb and Sn

56. An amalgam of mercury with sodium is categorized as a:
(A) Liquid in solid solution (B) Solid in liquid solution
(C) Liquid in liquid solution (D) Gas in solid solution

57. The uptake of hydrogen gas by palladium metal is an example of a:
(A) Gas in solid solution (B) Solid in gas solution
(C) Gas in liquid solution (D) Liquid in gas solution

58. A homogeneous mixture of camphor in nitrogen gas is an example of a:
(A) Solid in gas solution (B) Gas in solid solution
(C) Liquid in gas solution (D) Solid in liquid solution

59. What concentration of fluoride ions in drinking water causes tooth enamel to become mottled?
(A) 1.0 ppm (B) 1.5 ppm (C) 0.5 ppm (D) 2.0 ppm

60. Which chemical compound is commonly used as rat poison?
(A) $NaCl$ (B) NaF (C) KCl (D) $CaCl_2$

61. If the dissolution of a solid in a liquid is endothermic ($\Delta_{sol}H > 0$), raising the temperature causes solubility to:
(A) Increase (B) Decrease
(C) Remain constant (D) First decrease then increase

62. If the dissolution process of a solute is exothermic ($\Delta_{sol}H < 0$), raising the temperature causes solubility to:
(A) Increase (B) Decrease
(C) Remain unaffected (D) Become infinite

63. Pressure has a negligible effect on the solubility of solids in liquids because:
(A) Solids and liquids are practically incompressible
(B) Solids have extremely low density
(C) Liquids are highly volatile
(D) Pressure increases intermolecular distances

64. Henry's law holds most accurately under which of the following conditions?
(A) Pressure is excessively high
(B) The gas does not react chemically with the solvent
(C) Temperature is extremely low
(D) The solution is highly concentrated

65. Raoult's law becomes a special case of Henry's law when:
(A) $K_H = p_1^\circ$ (B) $K_H = 0$
(C) $p_1^\circ = 0$ (D) The solute is non-volatile

66. In the vapour phase, the partial pressure of component 1 is related to its vapour mole fraction (y_1) by:
(A) $p_1 = y_1 P_{total}$ (B) $p_1 = \frac{y_1}{P_{total}}$ (C) $p_1 = y_1 + P_{total}$ (D) $P_{total} = y_1 p_1$

67. Which of the following statements is strictly true for an ideal solution?
(A) It obeys Raoult's law across all concentrations (B) Heat is absorbed during its formation
(C) Its total volume decreases (D) It forms an azeotropic mixture

68. A mixture of carbon disulphide (CS_2) and acetone exhibits:
(A) Positive deviation from Raoult's law (B) Negative deviation from Raoult's law
(C) Completely ideal behavior (D) Complete insolubility

69. What type of deviation from Raoult's law is exhibited by a mixture of phenol and aniline?
(A) Positive deviation (B) Negative deviation
(C) No deviation (D) First positive then negative

70. Colligative properties of a solution depend fundamentally on the:
(A) Chemical nature of solute particles
(B) Total number of solute particles
(C) Chemical reactivity of solvent
(D) Color and appearance of the solution

71. The value of the molal depression constant (K_f) depends exclusively on the:
(A) Nature of the solute (B) Nature of the solvent
(C) Molality of the solution (D) Prevailing atmospheric pressure

72. The cryoscopic constant is another name for the:
(A) Molal elevation constant (B) Molal depression constant
(C) Henry's law constant (D) van 't Hoff factor

73. The ebullioscopic constant is directly associated with which colligative property?
(A) Elevation of boiling point (B) Depression of freezing point
(C) Osmotic pressure (D) Lowering of vapour pressure

74. If T_b° is the boiling point of pure solvent and T_b is that of solution, elevation of boiling point (ΔT_b) is:
(A) $T_b - T_b^\circ$ (B) $T_b^\circ - T_b$ (C) $T_b + T_b^\circ$ (D) $T_b \times T_b^\circ$

75. If T_f° is the freezing point of pure solvent and T_f is that of solution, depression of freezing point (ΔT_f) is:
(A) $T_f - T_f^\circ$ (B) $T_f^\circ - T_f$ (C) $T_f + T_f^\circ$ (D) $T_f^\circ \times T_f$

76. An aqueous solution freezes below 0 °C because adding a non-volatile solute causes:
(A) Vapour pressure to decrease (B) Vapour pressure to increase
(C) Surface tension to decrease (D) Solvent volume to expand

77. The boiling point of a liquid is the temperature at which its equilibrium vapour pressure equals:
(A) Zero (B) Atmospheric pressure
(C) 100 atmospheres (D) Vapour pressure at freezing point

78. An artificial semipermeable membrane is commonly prepared using:
(A) Copper sulphate (B) Copper ferrocyanide (C) Potassium ferrocyanide (D) Sodium chloride

79. When red blood cells are suspended in pure distilled water, they will:
(A) Shrivel (B) Swell and burst
(C) Remain intact (D) First shrink then expand

80. Salting meat and adding excess sugar to fruits preserves them from bacterial decay because:
(A) Bacteria directly feed on salt
(B) Bacteria lose water via osmosis, shrivel and die
(C) Salt is chemically poisonous to bacteria
(D) Temperature increases dramatically

81. Wilted flowers revive when placed in pure water due to which process?
(A) Diffusion (B) Endosmosis (C) Exosmosis (D) Transpiration

82. A 0.1 M urea solution and a 0.1 M glucose solution are mutually:
(A) Isotonic (B) Hypertonic
(C) Hypotonic (D) Azeotropic

83. If a solution possesses higher osmotic pressure relative to another solution, it is called:
(A) Hypotonic (B) Hypertonic
(C) Isotonic (D) Saturated

84. When solute molecules undergo association in a solvent, their experimental molar mass is:
(A) Lower than expected (B) Higher than expected
(C) Equal to expected (D) Zero

85. When solute molecules dissociate into ions in a solution, their observed molar mass is:
(A) Lower than normal (B) Higher than normal
(C) Unaltered (D) Infinite

86. The van 't Hoff factor (i) is defined as:
(A) Normal molar mass / Abnormal molar mass (B) Abnormal molar mass / Normal molar mass
(C) Calculated colligative property / Observed property (D) Moles of solute / Moles of solvent

87. The total number of ions produced by the complete ionization of 1 mole of $Al_2(SO_4)_3$ is:
(A) 2 (B) 3 (C) 5 (D) 4

88. At 0.01 M concentration, which of the following aqueous solutions will have the lowest freezing point?
(A) Glucose (B) Urea (C) $NaCl$ (D) $Al_2(SO_4)_3$

89. Which of the following 0.1 M aqueous solutions will exhibit the highest boiling point?
(A) $NaCl$ (B) $BaCl_2$ (C) $FeCl_3$ (D) Sucrose

90. Sprinkling salt ($NaCl$) onto icy roads causes ice to melt because salt:
(A) Increases the melting point of ice (B) Lowers the freezing point of water
(C) Exothermically generates heat (D) Enhances the vapour pressure of ice

91. The primary bonding interaction responsible for the dimerization of ethanoic acid in benzene is:
(A) Ionic bond (B) Hydrogen bond (C) Covalent bond (D) Metallic bond

92. For a weak electrolyte, the degree of dissociation (α) and van 't Hoff factor (i) are related by:
(A) $\alpha = \frac{i-1}{n-1}$ (B) $\alpha = \frac{n-1}{i-1}$ (C) $\alpha = \frac{i+1}{n+1}$ (D) $\alpha = i(n-1)$

93. The degree of association (x) and van 't Hoff factor (i) are related by (where n is the association number):
(A) $x = \frac{1-i}{1-1/n}$ (B) $x = \frac{i-1}{n-1}$ (C) $x = \frac{1-i}{n-1}$ (D) $x = \frac{i-1}{1-1/n}$

94. A 20% (w/w) aqueous KI solution indicates that 100 g of the solution contains:
(A) 20 g KI and 100 g water (B) 20 g KI and 80 g water
(C) 20 g water and 80 g KI (D) 20 g KI and 80 mL alcohol

95. A commercial liquid bleaching solution contains what mass percentage (w/w) of sodium hypochlorite in water?
(A) 10% (B) 3.62% (C) 5.8% (D) 0.9%

96. Aniline and chloroform cannot be separated by simple fractional distillation because:
(A) They are completely immiscible
(B) They form an azeotropic mixture
(C) They undergo violent chemical reaction
(D) They possess identical densities

97. When the temperature of a liquid is raised, its vapour pressure:
(A) Increases (B) Decreases
(C) Remains constant (D) First decreases then increases

98. Which of the following solutions will exhibit the lowest vapour pressure?
(A) 0.1 M glucose (B) 0.1 M $NaCl$ (C) 0.1 M $BaCl_2$ (D) 0.1 M $FeCl_3$

99. Elevation of boiling point for a dilute solution of a non-volatile solute is directly proportional to:
(A) Molarity (B) Molality (C) Volume percentage (D) Normality

100. The expression representing the relative lowering of vapour pressure for a non-volatile solute is:
(A) $\frac{p_1^\circ - p_1}{p_1^\circ} = x_2$ (B) $\frac{p_1 - p_1^\circ}{p_1^\circ} = x_1$ (C) $\frac{p_1^\circ - p_1}{p_1} = x_2$ (D) $p_1 = p_1^\circ x_2$